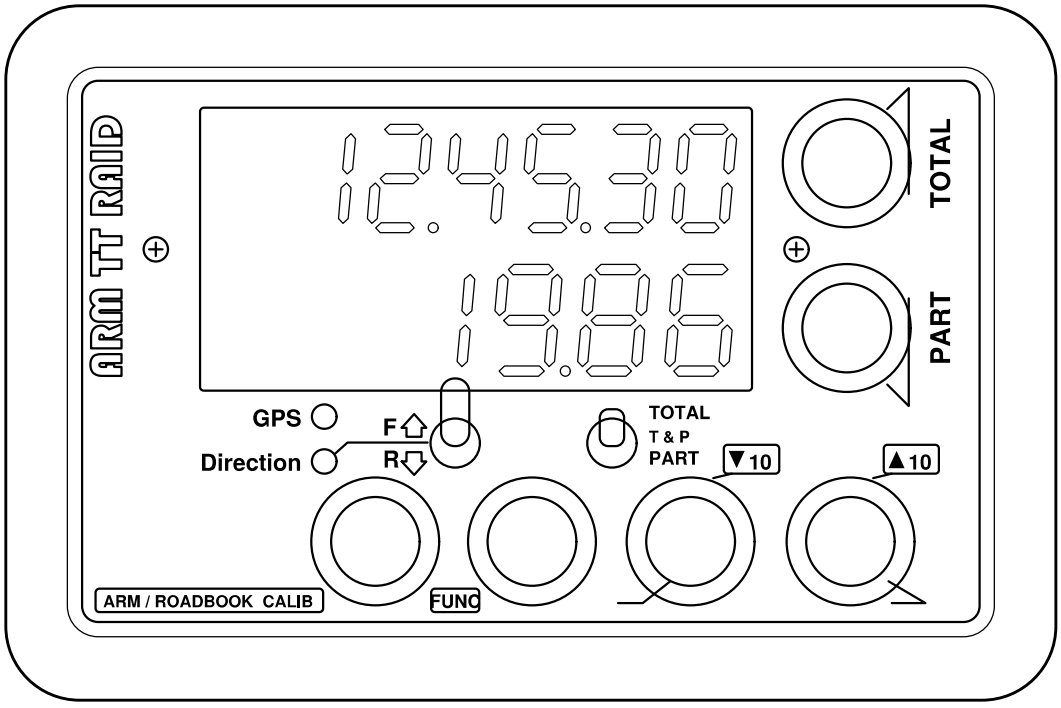
ARM products / GPS Trip Meter for Rally Raid

ARM TT Raid

User Manual

Please read this manual before use and keep it for future reference.



Contents

01	Safety precautions
02	Parts
03	Power and GPS
04	Switching the display
05	TOTAL and PART
06	±10 m buttons
07	Direction switch and trip selector
08	Distance calibration
09	Stage timing
10	Settings (brightness, time offset, ±10 m target)
11	Driver display (optional)
12	Remote switch
13	Sides and rear (connectors, mounting)
14	Specifications
15	Quick reference

CHAPTER 01

Safety precautions

▲ Power supply
Power the unit from **either USB-C (5 V) or DC12V, never both**. Supplying power from both at the same time may damage the unit.

▲ Operating while driving
All operation while driving should be done by the co-driver. The driver must concentrate on driving; mount the driver display where it can be read with minimal eye movement.

▲ GPS start-up time
The first time you use the unit, or after it has been switched off for a long time, it may take several minutes before GPS measurement starts. Switch it on well before the event or drive.

GPS reception
Distance and speed are measured by GPS. Reception may not be possible under a roof or in tunnels. After switching on, check GPS reception (GPS LED steady) in an open area before use.

Mounting location
Do not store the unit where water can pour on it directly or where it gets hot (e.g. in direct sunlight on the dashboard).

CHAPTER 02

Parts



No.	Name	Function
1	TOTAL button (white)	Short: resets TOTAL. Long: set TOTAL to any distance. Increases the value while setting.
2	PART button (white)	Short: resets PART. Long: set PART to any distance. Decreases the value while setting.
3	Red button (CALIB)	Short: stage timing (stage screen). Long: distance calibration. In the settings, a short press returns to the previous screen.
4	Blue button (FUNC)	Short: switches the upper display. Long: settings (brightness, time offset, ±10 m target).
5	Yellow button (left, ▼10)	−10 m; hold to repeat. Moves to the previous digit while calibrating or entering a distance.
6	Yellow button (right, ▲10)	+10 m; hold to repeat. Moves to the next digit while calibrating or entering a distance.
7	Upper display	TOTAL, clock, heading, speed or stage time.
8	Lower display	PART.
9	GPS LED (green)	Blinking: searching. Steady: GPS received.
10	Direction LED (white)	Off: forward. On: stop. Blinking: reverse.
11	Direction switch	Up: forward (F). Centre: stop. Down: reverse (R).

12	Trip selector	Up: TOTAL only. Centre: TOTAL and PART (T&P). Down: PART only.

Left side: USB-C (5 V) power at the top.

Right side: slide switch for DC12V at the top, with GX12 connectors below it (5-pin: driver display, 4-pin: remote switch, 3-pin: DC12V power).

Rear: mounting insert nuts (chapter 13).

CHAPTER 03

Power and GPS

Using USB-C

1. Connect 5 V to the USB-C port at the top of the left side. The unit switches on automatically.
2. While the green GPS LED blinks, the unit is searching for GPS. When it stays on, GPS is received.

Using DC12V

1. Connect DC12V (red) and GND (black) of the power cable to the 3-pin GX12 connector on the right side.
2. Slide the switch at the top of the right side down for ON, up for OFF.
3. The switch is not used when powering from USB-C.

What is kept
TOTAL and PART distances (saved every 10 m), calibration, brightness, time offset and the ±10 m target are kept when the power is off. Up to the last 10 m driven before switching off may not be restored. The stage time is not kept.

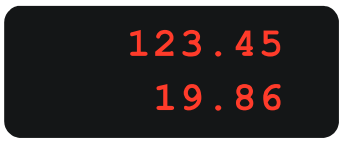
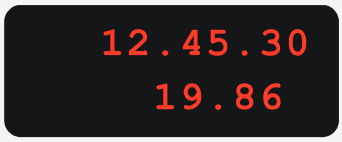
▲ Caution
Do not connect USB-C and DC12V at the same time.

Switch the power on while holding the red button to show the firmware version for 2 seconds.

CHAPTER 04

Switching the display

Each short press of the blue button switches the upper display as below. The lower display always shows PART.

Screen	Upper display	Example
TOTAL	Total distance (km, 10 m steps)	
Clock	GPS time (h.mm.ss)	
Heading	“H” followed by the heading in degrees (north = 0). “---” when stopped	
Speed	“SP” followed by the calibrated speed (km/h)	
Stage	Stage timing (chapter 9). “------” while on standby	

After the stage screen, the display returns to TOTAL.

CHAPTER 05

TOTAL and PART

Reset

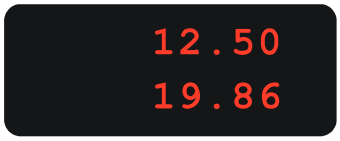
A short press of the TOTAL button resets TOTAL to 0, and a short press of the PART button resets PART. TOTAL is reset even while the upper display shows the clock or speed.

Setting any distance to measure from

1. Long press the TOTAL button to enter TOTAL (for PART, long press the PART button).
2. The TOTAL button adds 1 to the blinking digit, the PART button subtracts 1. The right yellow button moves to the next digit, the left one to the previous digit.
3. Long press the same button again (TOTAL button for TOTAL) to confirm.



Entering TOTAL



Set to 12.50 km

CHAPTER 06

±10 m buttons

Correct the distance against the road book quickly, in 10 m steps.

Button	Short press	Hold
Yellow (right, ▲10)	+10 m	Keeps increasing
Yellow (left, ▼10)	−10 m (no change below 10 m)	Keeps decreasing

Which trip is corrected (TOTAL and PART, TOTAL only or PART only) is chosen with the “±10 m target” setting (chapter 10). Both are corrected by default.

CHAPTER 07

Direction switch and trip selector

Direction switch

Switch	Action	Direction LED
Up (F)	Distance increases.	Off
Centre	Distance stops.	On
Down (R)	Distance decreases (e.g. to recover after a wrong turn).	Blinking

Trip selector

Switch	Trips that count distance
Up (TOTAL)	TOTAL only
Centre (T & P)	Both TOTAL and PART
Down (PART)	PART only

The two switches work together. For example, “PART only” with reverse decreases only PART.

CHAPTER 08

Distance calibration

When the displayed distance does not match the road book, adjust it with the calibration factor.

Example: a section of 10.00 km in the road book shows 10.20 km on TOTAL.

1. Long press the red button to enter calibration. The upper display shows “CALIb”, the lower display the factor.
2. The TOTAL button adds 1 to the blinking digit, the PART button subtracts 1. The right yellow button moves to the next digit, the left one to the previous digit.
3. Set 10.20 ÷ 10.00 = 1.0200. Each change is saved immediately.
4. Long press the red button again to return.

How to calculate
Calibration factor = ARM displayed distance ÷ road book distance



Calibration screen



Set to 1.0200

The factor also applies to the speed display.

CHAPTER 09

Stage timing

1. Switch the upper display to the stage screen (“------”) with the blue button.
2. A short press of the red button starts a countdown to 00 s of the next minute. TOTAL and PART are reset to 0.
3. At 00 s the stage starts automatically. Press the red button during the countdown to start immediately, or the blue button to cancel.
4. While running, the elapsed time (h.mm.ss) is shown with blinking separators.
5. Press the red button at the finish. The time blinks and PART returns to 0.
6. Press the red button again to return to standby (“------”).



Countdown (seconds left)



Running



Finish

The stage time is lost when the power is off.

CHAPTER 10

Settings (brightness, time offset, ±10 m target)

Long press the blue button to enter the settings; each short press of the blue button moves to the next setting. The last press returns to the previous screen. In any setting, a short press of the red button returns at once.

Order	Upper display	Setting	Operation
1	ILL	Brightness (Hi / Lo). The lower display shows Hi or Lo.	TOTAL / PART: switch
2	Utc	Time offset from UTC (−15 to +15), e.g. UK 0, France +1 (winter), Japan +9.	TOTAL: +1 PART: −1
3	10	±10 m target (“Func” on the lower display). The row with the “o” at the right end is the target: both rows = TOTAL and PART, upper only = TOTAL, lower only = PART.	TOTAL / PART: switch



Brightness



Time offset



±10 m target (both)

The settings are kept when the power is off.

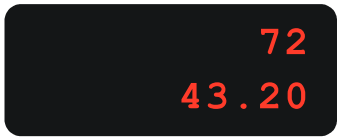
CHAPTER 11

Driver display (optional)

The driver display is an optional accessory.

Connect a driver display to the 5-pin GX12 connector on the right side to show information in front of the driver. It shows the same layout as the 2-row driver display of the ARM Twin Trip GPS2027.

Row	Display
Upper	Speed (km/h, calibrated)
Lower	PART (km)



Driver display

CHAPTER 12

Remote switch

Connect the remote switch (1 button) to the 4-pin GX12 connector on the right side.

Button	Function
Remote switch	Resets PART (short or long press, on the normal screens)

Mount it near the steering wheel or gear lever to reset PART at each road book box without reaching for the unit.

CHAPTER 13

Sides and rear (connectors, mounting)

Sides

Location	Part	Use
Left side, top	USB-C	5 V power
Right side, top	Slide switch	DC12V power ON / OFF (down = ON)
Right side	GX12 connector (5-pin)	Driver display
Right side	GX12 connector (4-pin)	Remote switch
Right side	GX12 connector (3-pin)	DC12V power (red: DC12V, black: GND)

Mounting insert nuts (rear)

Two mounting nuts are embedded in the rear, so the unit can be bolted to a bracket or mount.

Item	Details
Bolt	M4
Bolt length	Up to the thickness of the mounting part (bracket etc.) + 10 mm
Nut spacing	49 mm (centre to centre). RAM Mount bases with the same spacing can be used.

▲ Caution
A bolt that is too long may damage the inside of the unit. Always use bolts no longer than “thickness + 10 mm”.

CHAPTER 14

Specifications

Item	Details
Dimensions	W 145 × H 96 × D 27 mm (excluding buttons, switches, connectors and other protrusions)
Power	USB-C (5 V, left side) or DC12V (3-pin GX12 on the right side, with slide switch). Do not use both at the same time.
Distance / speed	Measured by GPS. Calibration factor 0.0000–2.9999
Display	Upper 6 digits (TOTAL, clock, heading, speed, stage time), lower 6 digits (PART)
Memory	TOTAL and PART distances (every 10 m), calibration, brightness, time offset, ±10 m target
Connections	GX12 connectors: driver display (5-pin), remote switch (4-pin), DC12V power (3-pin)
Mounting	Two M4 insert nuts on the rear (49 mm pitch)
Options	Driver display

CHAPTER 15

Quick reference

Button	Short press	Long press
TOTAL (white)	Reset TOTAL	Set TOTAL to any distance (confirm with a long press)
PART (white)	Reset PART	Set PART to any distance (confirm with a long press)
Red (CALIB)	Stage screen: countdown → start → finish → standby	Distance calibration (long press again to return)
Blue (FUNC)	Switch the upper display / cancel the countdown	Settings (brightness → time offset → ±10 m target)
Yellow (right, ▲10)	+10 m	Keeps increasing
Yellow (left, ▼10)	−10 m	Keeps decreasing
Remote switch	Reset PART	Reset PART

Screen	TOTAL (white)	PART (white)	Yellow right / left	Back
Calibration (CALIb)	+1	−1	Next / previous digit	Red long press
Distance entry	+1	−1	Next / previous digit	Long press of the same button
ILL / Utc / 10	+ / switch	− / switch	—	Last blue press or red short press

An animated manual is also available: manual.rally-arm.com